

Cresca – Comprehensive Product Blueprint

Integrated vision, features, tech, monetisation & roadmap
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1. Vision & Differentiators

Cresca is a cross-platform music ecosystem blending hi-fidelity streaming, hyperpersonalised discovery, community social layers, and creator-first monetisation. It differentiates via: - Magic-feeling on-device + cloud AI personalisation - Best-in-class audio (hi-res, spatial, personal HRTF) - Viral hooks (live sessions, shareable clips, countdown drops) - Empowered artists (stems marketplace, direct fan chat) - Deep accessibility (haptic cues, voice navigation)

2. Core & Foundational Features

2.1 Catalogue & Licensing

- Major-label + indie licensing (Sony, Warner, UMG, Merlin, Believe, DistroKid).
- Podcasts, audiobooks, live sets; DDEX ingestion with real-time rights checks.

2.2 Playback & Offline

- Adaptive streaming 128-256 kbps AAC → lossless FLAC.
- Rust engine ensures gapless & crossfade; Smart Downloads predict caching.

2.3 Accounts, Search & Playlists

- Email/social auth, profiles, stats; Elasticsearch deep search.
- Collaborative/private playlists; robust queue controls.

3. Next-Gen Personalisation & Discovery

3.1 AI Concierge DJ ★

- Conversational voice/text DJ stitching tracks & trivia.
- NLP prompt → PyTorch ranking; on-device Core ML for low-latency voice.

3.2 Prompt-to-Playlist

- Natural language playlist generator; GPT prompt vector → CF candidates.

3.3 On-Device Sentiment Model ★

- Sensor-driven mood detection (focus, cardio, relax).
- Privacy-first inference on Android NN / Core ML.

4. Audiophile Cred & Immersion

4.1 Adaptive Hi-Res Switching

- Network-aware FLAC 48-192 kHz; 'Hi-Res' badge in player.

4.2 Spatial/Atmos + Upmixer ★

- Native Dolby Atmos; ML stereo upmix to 3D image.

4.3 Personal HRTF Calibration ★

- Ear-scan selfie → nearest HRTF; synced across devices.

5. Social & Virality

5.1 Live Group Sessions

- WebRTC sync, mic■chat, emoji; host DJ hand■off.

5.2 1■Tap Social Clips

- FFmpeg 15 s clip, watermarked deep■link, TikTok API.

5.3 Countdown Drops ★

- Shareable timers auto■add releases; FOMO driver.

6. Creator & Indie Toolkit

6.1 Self■Service Upload + Analytics

- Drag■drop upload; real■time dashboards via WebSockets.

6.2 Stem Marketplace ★

- Individual stems for remix; Solana low■fee payments; Cresca 10 %.

6.3 Direct■to■Fan Chat

- Voice/text inbox; ticket codes; moderated.

7. Contextual Utility

7.1 Noise■Adaptive Cross■Fade ★

- Mic detects chatter; raises volume or cross■fades smoothly.

7.2 Offline Route Cache

- Route + coverage map predicts download needs.

7.3 Wellness Hub

- Binaural beats, journaling prompts; streak tracked to HealthKit/Fit.

8. Accessibility & Inclusivity

- Dynamic lyric fonts, dyslexia mode, high■contrast themes.
- Haptic beat cues; intensity slider.
- Voice■only navigation via shared NLP intents.

9. Additional Features

- Karaoke scoring; shareable performance clips.
- Timestamp sharing of any track/podcast moment.
- In■app concerts & Cresca Live ticketing.
- Music trivia games; AI beat■making sandbox.

10. Technical Architecture

10.1 Client Layer

- Flutter (mobile/desktop), Next.js (web).

10.2 Audio Engine

- Rust core, WebAssembly for web playback, Atmos decoding.

10.3 Backend & Data

- Node/Express GraphQL, MongoDB, Elasticsearch, Redis, S3.

10.4 Realtime & ML

- WebRTC, MQTT; PyTorch recommender, on-device ML for sentiment/HRTF.

10.5 Rights & Security

- DDEX, fingerprinting, Widevine/FairPlay DRM, zero-trust infra.

11. Monetisation Levers

- Freemium (ads, 160 kbps) → Premium → Hi-Fi tier.
- Stems marketplace & virtual merch (10 % fee).
- Event layer (tickets, AR posts).
- Family+Friends plan (1 + 4).
- NFT themes & exclusives.

12. Rollout Roadmap

- MVP (0–9 m): catalogue, streaming, Prompt playlist, offline, lyrics.
- Phase 2 (9–18 m): AI DJ, Live Sessions, social clips, accessibility.
- Phase 3 (18–24 m): Creator tools, stems, countdown drops, concerts.
- Phase 4 (24 m+): Spatial/HRTF, noise crossfade, wellness hub.

13. Competitive Snapshot

Feature	Spotify	YouTube Music	Apple Music	Cresca
AI DJ	Partial	No	No	Full
Spatial + Personal HRTF	Limited	No	Spatial only	Yes
Stems Marketplace	No	No	No	Yes
NoiseAdaptive Fade	No	No	No	Yes
Live Group Chat	Jam (text)	No	No	Yes
On-device Sentiment	No	No	No	Yes

14. Strategic Guardrails & Next Steps

- Licensing & DDEX pipeline – immediate priority.
- Data-led iteration; A/B test AI DJ & Sentiment features.
- WCAG 2.2 accessibility baseline.
- Kubernetes + multi-CDN for global latency targets.
- Prepare UX wireflows (AI DJ, Creator Dash) and legal rights map.